

## HS 545: Applied Forensic Psychology

|                  |                                                                                                                                                                                   |
|------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Course Code      | HS 545                                                                                                                                                                            |
| Course Name      | Applied Forensic Psychology                                                                                                                                                       |
| L-T-P-C          | 3-0-2-4                                                                                                                                                                           |
| Prerequisites    | Basic Knowledge of Psychology and cognitive processes (Advised to enroll for Introduction to Psychology (HS252) course before joining this course), or Consent of the Instructor. |
| Intended for     | B.Tech/Masters/PhD                                                                                                                                                                |
| Mutual Exclusion | None                                                                                                                                                                              |
| Approval         | 45th BoA (Feb 2022)                                                                                                                                                               |

### Course Contents

- Module 1: Overview of Forensic and Criminal Psychology (8 hours)
  - Forensic Psychology: Major areas, Scope, State of art
  - Role of Forensic psychologist: Educational and certification requirements
  - Theories of crime:
    - Lombroso approach to understand crime, its causes and remedies
    - Moral reasoning Theory
    - Social information-processing Theory
    - Developmental and psychological theories of offendin
    - Developmental propensity theory
    - Propensity Theory
    - Social learning Theory
    - Lifestyle Theory
    - Integrated Cognitive Antisocial Potential theory
  - Recent Psychological research and police investigations: Does the research meet the needs? Victims of crime
- Module 2: Eyewitness testimony (6 hours)
  - Eyewitness testimony as a central issue in criminal and forensic psychology
  - Accuracy of the witness evidence: Role of System and Estimator variables
  - Eyewitness evidence in courts
  - Children as witnesses
  - Factors that lead to criminal behaviour, criminal profiling
  - Scope for psychological assessments of suspects for facilitating understanding of the crime committed
- Module 3: Witness memory: encoding, storage and retrieval factors (6 hours)
  - Mistaken identity
  - Perceiving event
  - Retaining information in memory and retrieving the same Inattentional blindness
- Module 4: Visual Identification (6 hours)
  - Perceiving events
  - Recognizing people and facial recognition
  - Individual differences in eyewitness testimony
- Module 5: Profile Analysis: (6 hours)
  - The origin of offender's profile
  - The process of police investigation
  - Type of profiling: Inductive and deductive
  - Profiling and personality
  - What research speaks about profiling: whether it works?
- Module 6: Lies, Lie detecting and act of deception: Technological and non-technological involvement (6 hours)
  - Brain development and the parts of the brain underpinning social and antisocial behaviours

- Ekman's theory of lie detection
- Improving lie detection hit rates: Cognitive overload
- The strategic use of evidence technique
  - \* The polygraph process and its validity, Alternatives to the polygraph
  - \* Layered Voice Analysis (LVA)
  - \* Suspect Detection System (SDS)
  - \* Brain Electrical Oscillations Signature (BEOS)
  - \* Gait analysis o Handwriting analysis
  - \* Use of Eye tracker and electroencephalograph in investigations
  - \* Non-technological forensic assessments/investigations: MMPI, PCL-R, TAT, Expert testimony
- Module 7: Interviewing/interrogating witnesses and suspects (8 hours)
  - Methods of interviewing witnesses and suspects:
  - Cognitive interview
  - Interviewing assailable witnesses such as children, elderly people and people with intellectual disability
  - Working with the courts: Judicial processes
  - Presentation of evidences in courts
  - Judges as decision makers
  - Advice for expert witnesses
  - Case studies of criminals with concluding judgments and convictions

#### Laboratory/practical/tutorial Modules: (10 hours)

Laboratory and practical sessions will be distributed into different modules (module, 2, 3, 4, 5 and 6). Students will be introduced to some of the forensic assessment tools, such as Rorschach Ink Blot, Personality Big-5, Psychopathic Checklist-Revised (PSL-R), Minnesota Multiple Personality Inventory-3, Electroencephalography, Thematic Apperception Test, Galvenic skin conductance etc. Additionally, expert testimony will also be included in the syllabus.

- Part 1: Personality test introduction and administration (paper-pencil based)
- Part 2: Electroencephalography and Galvanic skin conductance/lie detection tools introduction and administration
- Part 3: Field work, visiting police stations, courts interacting with police personnel and people in judiciary system to know the nature of criminal investigation and proceedings

#### Text Books

1. Graham. M. Davies and Anthony R. Beech, **Forensic psychology: Crime, justice, law, interventions**, 3rd Edition, John Wiley & Sons, 2018.
2. Curt R. Bartol and Anne M. Bartol, **Introduction to Forensic Psychology**, 5th Edition, Sage Publication, Thousand Oaks, California, 2018.

#### References:

1. Mickes, L. (2015). Receiver operating characteristic analysis and confidence-accuracy characteristic analysis in investigations of system variables and estimator variables that affect eyewitness memory. *Journal of Applied Research in Memory and Cognition*, 4, 93-102.
2. Sridhar Ramamurty, David E. Morrison III, Joseph W. Koletar and Kelly R. Pope, **A.B.C.'s of Behavioral Forensics: Applying Psychology to financial fraud and detection**, John Wiley & Sons, Hoboken, New Jersey, 2013.
3. Philip H. Melanson, **The Murkin Conspiracy: An Investigation into the Assassination of Dr. Martin Luther King Jr.**, Praeger Publication, 1989.
4. Dennis Howitt, **Introduction to forensic and criminal psychology**, 6th Edition, Pearson Education, 2006.
5. Lombroso, C., **Crime, its causes and remedies** (Vol. 3). The University Press, 1911.
6. Elizabeth F. Loftus, **Eyewitness Testimony**, Harvard University Press, 1979.